RETAILPULSE

AI-Powered Customer Analytics & Demand Forecasting Platform

PROJECT REPORT

An end-to-end retail analytics demonstration combining Python-based analysis, customer segmentation, demand forecasting, inventory planning and Power BI reporting.

Project area	Implementation
Data analysis	Python, Pandas, NumPy
Customer analytics	RFM scoring and rule-based customer segments
Churn analysis	Baseline and class-weighted XGBoost experiments
Demand forecasting	Moving Average, Croston-SBA and Prophet comparison
Inventory planning	Forecast-based lead-time demand and reorder estimates
Business intelligence	Power BI report with 12 planned analytical pages

Prepared by: Hariom Upadhyay

Data note: This project uses a synthetic dataset created for analysis and demonstration. It does not represent actual transactions from a real retailer.

1. Executive Summary

RetailPulse demonstrates how retail transaction, customer, product, demand and inventory data can be brought together to support business reporting and operational decisions. The workflow covers data inspection, cleaning checks, exploratory analysis, customer segmentation, churn experiments, intermittent-demand forecasting, inventory reorder estimation and Power BI dashboard design.

The synthetic dataset contains 30,000 transaction records, 1,500 customers, 120 products and approximately two years of daily product-demand observations. The analysis identifies sales and profitability patterns, groups customers using RFM scores, compares forecasting approaches and estimates replenishment needs using forecast demand during supplier lead time.

The work is a portfolio demonstration rather than a production deployment. In particular, the churn model did not demonstrate useful predictive performance, and the demand-forecasting evaluation did not meet the original 12% MAPE target. These limitations are reported transparently.

2. Project Objectives

• Build a structured retail analytics workflow using multiple related datasets.
• Calculate core business KPIs and analyze sales by category and month.
• Segment customers using Recency, Frequency and Monetary value.
• Evaluate a baseline churn-classification approach and identify data-leakage risks.
• Compare forecasting methods for sparse/intermittent product demand.
• Estimate inventory reorder quantities using forecast lead-time demand and safety stock.
• Present business insights through a multi-page Power BI report.

3. Dataset Overview

Dataset	Rows	Purpose
customers.csv	1,500	Customer profile and signup information
products.csv	120	Product and category attributes
sales_transactions.csv	30,000	Transaction-level sales, revenue, profit and quantity
customer_features_churn.csv	1,500	Customer features and churn-status label
inventory_snapshot.csv	120	Stock, reorder level, lead time and safety stock
daily_product_demand.csv	25,453	Daily product demand observations

Data quality checks found no reported missing values or duplicate rows in the initial inspection. Date columns were converted to datetime format and checked for invalid values. The sales and demand history covers 1 January 2024 to 30 December 2025; the inventory snapshot is dated 31 December 2025.

Dataset relationships: Customer_ID links customer and transaction/customer-feature tables. Product_ID links products, transactions, inventory and daily demand. The six datasets are intentionally kept separate and joined only when required for a specific analysis.

4. Business KPI Analysis

KPI	Result
Total Revenue	■95,526,081.29 (approximately ■9.55 crore)
Total Profit	■29,266,222.12 (approximately ■2.93 crore)
Distinct Orders	30,000
Total Quantity Sold	52,435
Average Order Value	■3,184.20
Weighted Profit Margin	30.64%

Average Order Value is calculated as total revenue divided by distinct order count. Overall profit margin is calculated as total profit divided by total revenue; this is a revenue-weighted overall margin, not the simple average of row-level margins.

5. Category Performance

Category	Revenue (■)	Profit (■)	Units	Margin
Grocery	19,217,210.54	5,197,911.58	9,763	27.05%
Fashion	18,597,415.70	6,151,876.62	9,665	33.08%
Electronics	17,289,782.35	5,144,964.01	10,439	29.76%
Sports	15,439,993.85	4,793,886.26	7,749	31.05%
Home & Kitchen	12,685,580.89	4,070,572.22	6,669	32.09%
Beauty	12,296,097.96	3,907,011.43	8,150	31.77%

Key observations: Grocery generated the highest revenue; Fashion generated the highest profit and category margin; Electronics recorded the highest quantity sold. Beauty had the lowest revenue, while its margin remained above the overall margin.

6. Monthly Sales Analysis

Monthly revenue was analyzed across 24 months. July 2025 recorded the highest monthly revenue at ■4,360,636.24, with 1,340 orders. February 2025 recorded the lowest monthly revenue at ■3,747,905.99. Month-over-month growth was calculated using the previous month's revenue; the first month has no prior month for comparison.

The strongest reported month-over-month increase was +11.86% in July 2025, while the largest decline was -10.79% in February 2024. These results describe the synthetic dataset and should not be interpreted as real market seasonality.

7. Customer Segmentation (RFM)

Customers were scored from 1 to 5 on Recency, Frequency and Monetary value. Higher Recency scores represent more recent purchases; higher Frequency and Monetary scores represent more orders and spending. The three scores were combined to create RFM profiles and business-oriented segments.

Segment	Customers	Average Spending (■)	Segment Revenue (■)
At Risk	331	69,875.17	23,128,682.62
Champions	215	86,843.74	18,671,404.45

Segment	Customers	Average Spending (■)	Segment Revenue (■)
Hibernating	262	45,344.01	11,880,131.82
Loyal Customers	165	76,096.83	12,555,976.56
Needs Attention	238	54,309.82	12,925,736.61
New / Recent Customers	213	51,879.98	11,050,436.60
Potential Loyalists	76	69,917.27	5,313,712.63

At Risk was the largest segment with 331 customers (about 22.1%). Champions had the highest average spending at approximately ■86.8k. Segment rules are heuristic business definitions, not a learned clustering model.

8. Churn Analysis

The churn label in the supplied synthetic data is defined as no purchase for more than 90 days as of 31 December 2025. The target distribution is 1,383 active customers (92.2%) and 117 churned customers (7.8%), indicating class imbalance.

A baseline XGBoost classifier using non-leakage-prone customer features achieved 0.9233 accuracy but predicted no positive churn cases: precision, recall and F1 for the churn class were all zero, with ROC-AUC approximately 0.505. A class-weighted experiment did not improve useful predictive performance (ROC-AUC approximately 0.463; churn recall approximately 0.087).

A rule using Recency_Days greater than 90 reproduces the label with 100% accuracy because the label itself was defined using that condition. This is target leakage, not an independent predictive result. Recency_Days, Days_Since_Last_Purchase, Last_Purchase_Date and derived RFM fields must not be used to claim future churn prediction when they directly encode the label.

Conclusion: the current dataset supports a descriptive 90-day inactivity segment, but it does not establish a reliable forward-looking churn model. A stronger future design needs historical customer snapshots and labels based on purchases occurring after each snapshot date.

9. Demand Forecasting

Daily product demand is sparse and intermittent. For a representative product, 539 of 730 days (73.84%) had zero demand. Because many actual values are zero, ordinary MAPE is unsuitable or unstable; MAE, RMSE and WAPE were used for evaluation.

Model	Average MAE	Average RMSE	Average WAPE
Croston-SBA	0.855	1.155	156.705%
Moving Average	0.858	1.168	157.625%
Prophet	0.862	1.156	158.441%

Across 120 products and three rolling validation folds (1,080 model evaluations), Croston-SBA had the best average WAPE among the three tested methods, although the differences were small. The selected method by product was Prophet for 53 products, Croston-SBA for 34 and Moving Average for 33.

The original target of MAPE at or below 12% was not achieved. The reported WAPE values are high, indicating substantial forecast error. Forecasts should therefore be treated as exploratory planning estimates, not reliable production forecasts.

10. Forecast-Based Inventory Planning

Inventory planning combined forecast demand during supplier lead time with safety stock. The estimated reorder point was calculated as forecast lead-time demand plus safety stock. Recommended purchase quantity was the positive difference between the reorder point and current stock, rounded up to whole units.

Measure	Result
Products requiring reorder	12
Total recommended reorder units	268
Estimated purchase cost	■323,173.72 (approximately ■3.23 lakh)
Products with sufficient stock	108

Home & Kitchen had the largest estimated purchase budget (approximately ■103,150), while Sports had the highest recommended unit quantity (99 units). The estimate assumes no incoming purchase orders and uses a single inventory snapshot. Because forecast accuracy is weak, reorder recommendations require human review before any real purchasing decision.

11. Power BI Dashboard

The Power BI report is organized into 12 analytical pages to help users explore the retail data from executive-level KPIs through detailed operational views. The intended pages are:

1. Executive Summary
2. Sales Trends
3. Product Performance
4. Customer Analytics
5. Region & Category Analysis
6. Monthly & Seasonal Analysis
7. Customer Behaviour Analysis
8. Revenue & Profitability
9. Inventory Risk Analysis
10. Orders & Transactions Analysis
11. Advanced Analytics
12. Interactive Dashboard

The report uses KPI cards, line charts, bar/column charts, donut charts, scatter plots and slicers. A Date_Table supports year, month and date filtering. Page Navigator is an optional convenience for switching between report pages.

12. Limitations and Responsible Interpretation

• The dataset is synthetic and does not represent a real retailer.
• The churn classifier did not demonstrate useful predictive performance; the perfect recency rule is label leakage and must not be presented as a predictive model.
• The demand forecasting target of MAPE ≤12% was not achieved; WAPE results were high.
• Intermittent demand and many zero-demand days make percentage-based errors difficult to interpret.
• Inventory reorder quantities depend on forecast quality, lead-time assumptions, current stock snapshot and safety-stock inputs.
• The forecasting and inventory outputs are analytical prototypes, not production-grade recommendations.

13. Future Improvements

• Obtain real, time-stamped retail sales and inventory data, including promotions, prices, stockout periods and purchase orders.
• Build leakage-safe churn labels using time-based snapshots and future outcome windows.
• Test additional intermittent-demand and hierarchical forecasting methods; tune models with rolling-origin validation.
• Evaluate forecast bias and service-level performance in addition to MAE/RMSE/WAPE.
• Add incoming purchase orders, supplier constraints, minimum order quantities and service-level targets to inventory planning.
• Deploy a Streamlit interface and add model tracking/monitoring only after the analytical models are validated.

14. Conclusion

RetailPulse demonstrates an end-to-end analytics workflow: from data inspection and business KPI analysis to RFM segmentation, forecasting experiments, inventory planning and Power BI reporting. It produces clear descriptive insights and a structured dashboard while also documenting where predictive performance is not yet sufficient. The next stage is to validate the approach on real retail data and improve the churn and demand forecasting methodology before operational use.

End of Report